

AYMANE AIT DADS

Data Science Intern - Model Optimization | Quantization · Numerical Accuracy · Neural Network Inference

Ayman.Ait-dads@eurecom.fr | +33 7 60 92 50 93 | linkedin.com/in/aymane-ait-dads | Sophia Antipolis, France

PROFESSIONAL SUMMARY

Ranked 17th out of 3,000+ teams in the NVIDIA Nemotron Reasoning Challenge (score 0.86/1.0, \$106K+ prize pool) by architecting a full LoRA SFT pipeline on a 30B-parameter model with BF16 precision, completion-only loss, and 8,192-token context - directly demonstrating end-to-end neural network training and numerical precision expertise. Built and debugged a per-layer diagnostic evaluation pipeline revealing that hard-task failures stemmed from reasoning over-expansion rather than incorrect reasoning, reducing manual analysis time by 60% at Orange Maroc through automated feature engineering on 100K+ records. Brings solid Python, PyTorch, NumPy, and Matplotlib skills alongside hands-on experience with Transformers, CNNs, ViTs, and low-level precision formats (FP16/BF16) directly applicable to Quadric's quantization library and numerical validation tooling. Aims to contribute to calibration dataset construction, per-layer error visualization, and numerical accuracy testing infrastructure for Quadric's GPNPU edge inference architecture.

CORE COMPETENCIES

Quantization Workflows | Numerical Accuracy Testing | Per-Layer Error Debugging | Calibration Datasets | Model Optimization | CNN & Transformer Familiarity | PyTorch / NumPy | Fixed-Point & Mixed Precision

WORK EXPERIENCE

Orange Maroc

Jul 2024 - Aug 2024

Data Science Intern - Casablanca, Morocco

- Built and debugged a **Random Forest + K-Means pipeline** on 100K+ fiber-optic network records, classifying quality across thousands of nodes using upload speed, latency, and jitter features.
- Developed a clustering workflow mapping customer performance profiles to **5 commercial service tiers**, with outputs adopted directly by the network optimization team for maintenance decisions.
- Reduced manual analysis time by **~60%** through automated feature engineering, model evaluation scripts, and distribution visualization using Pandas, Scikit-learn, and Power BI.
- Delivered stakeholder presentations translating **model outputs into actionable maintenance recommendations**, bridging data science findings and non-technical business decision-makers.

PROJECTS

NVIDIA Nemotron Model Reasoning Challenge - Kaggle (In Progress)

Kaggle Competition · 2025-2026 · Ranked 17th / 3,000+ Teams

- Ranked **17th / 3,000+ teams** (score 0.86/1.0, \$106K+ prize pool) fine-tuning a 30B-parameter Mamba-Transformer with LoRA SFT ($r=32$, BF16, completion-only loss, 8,192-token context, $\text{grad_accum}=32$).
- Ran **controlled ablation experiments** across LR schedules, packing strategies, and synthetic augmentation - discovering reasoning-style consistency outweighs data volume; built custom cipher parsers and 8-bit operation solvers for verified chain-of-thought data.

PyTorch · Unsloth · TRL · PEFT · Hugging Face Transformers · vLLM · Triton · Kaggle GPU

AI-Generated Image Detection -

EURECOM ImSecu Course + NTIRE 2026 @ CVPR · 2025 · Ranked 1st in EURECOM Class (0.791 AUC)

NTIRE 2026 @ CVPR

- Ranked **1st in EURECOM class** (private Kaggle leaderboard, 0.791 AUC) fine-tuning CLIP ViT-L/14 (428M params) with a custom MLP head, dual learning rates, and FP16 mixed precision on 250K images across 25+ generator types.
- Submitted to **NTIRE 2026 @ CVPR CodaBench** international benchmark; found 1-epoch training (0.793 AUC) outperformed 4-epoch (0.767 AUC), demonstrating that early-stopping improves out-of-distribution generalization for vision Transformers.

PyTorch · OpenCLIP · ViT · AdamW · FP16 · Kaggle GPU

EDUCATION

Engineering Degree in Data Science (Master-Level) - EURECOM

Sep 2023 - Present

Relevant coursework: Machine Learning, Deep Learning, Statistics, Computer Vision, NLP, Image Security, Cloud Computing

CPGE - Mathematics & Physics - Ibn Timiya

2021 - 2023

SKILLS

ML Frameworks & Precision Training: PyTorch | NumPy | Scikit-learn | Hugging Face Transformers | FP16 / BF16 Mixed Precision | PEFT / LoRA | TRL SFTTrainer

Model Architecture & Optimization: CNN | ViT / Transformer | CLIP | DenseNet | Ablation Studies | Ensemble Methods | vLLM | Triton

Data Engineering & Visualization: Python | Pandas | Matplotlib | Seaborn | Power BI | SQL | Git | Linux